4.11.14

EE25BTECH11036 - M Chanakya Srinivas

Question

Find the value of λ for which the following lines are perpendicular to each other. Hence determine whether the lines intersect or not.

$$\frac{x-5}{5\lambda+2} = \frac{2-y}{5} = \frac{1-z}{-1}, \tag{1}$$

$$\frac{x}{1} = \frac{y+\frac{1}{2}}{2\lambda} = \frac{z-1}{3}. \tag{2}$$

Solution

From the symmetric form, the point and direction vectors of each line are

$$\mathbf{A} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{m_1} = \begin{pmatrix} 5\lambda+2 \\ -5 \\ 1 \end{pmatrix}, \tag{3}$$

$$\mathbf{B} = \begin{pmatrix} 0 \\ -\frac{1}{2} \\ 1 \end{pmatrix}, \qquad \mathbf{m_2} = \begin{pmatrix} 1 \\ 2\lambda \\ 3 \end{pmatrix}. \tag{4}$$

Perpendicular condition:

$$\begin{aligned} \mathbf{m_1}^\top \mathbf{m_2} &= 0, \\ (5\lambda+2)(1) + (-5)(2\lambda) + (1)(3) &= 0, \\ 5\lambda + 2 - 10\lambda + 3 &= 0, \\ -5\lambda + 5 &= 0, \\ \Rightarrow \boxed{\lambda = 1}. \end{aligned} \tag{5}$$

Substituting $\lambda = 1$,

$$\mathbf{m_1} = \begin{pmatrix} 7 \\ -5 \\ 1 \end{pmatrix}, \quad \mathbf{m_2} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}. \tag{6}$$

The matrix M and vector $\mathbf{B} - \mathbf{A}$ are

$$M = \begin{pmatrix} 7 & 1 \\ -5 & 2 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{B} - \mathbf{A} = \begin{pmatrix} -5 \\ -\frac{5}{2} \\ 0 \end{pmatrix}. \tag{7}$$

The lines intersect if

$$\operatorname{rank}\begin{pmatrix} M & \mathbf{B} - \mathbf{A} \end{pmatrix} = 2. \tag{8}$$

RREF Calculation

Form the augmented matrix:

$$\begin{pmatrix} 7 & 1 & -5 \\ -5 & 2 & -\frac{5}{2} \\ 1 & 3 & 0 \end{pmatrix} \tag{9}$$

Perform row operations:

$$R_2 \leftarrow R_2 + \tfrac{5}{7}R_1 = \begin{pmatrix} 7 & 1 & -5 \\ 0 & \frac{19}{7} & -\frac{75}{14} \\ 1 & 3 & 0 \end{pmatrix}, \tag{10}$$

$$R_3 \leftarrow R_3 - \tfrac{1}{7}R_1 = \begin{pmatrix} 7 & 1 & -5 \\ 0 & \frac{19}{7} & -\frac{75}{14} \\ 0 & \frac{20}{7} & \frac{5}{7} \end{pmatrix}. \tag{11}$$

Next,

$$R_2 \leftarrow \tfrac{7}{19}R_2 = \begin{pmatrix} 7 & 1 & -5 \\ 0 & 1 & -\frac{75}{38} \\ 0 & \frac{20}{7} & \frac{5}{7} \end{pmatrix}, \tag{12}$$

$$R_3 \leftarrow R_3 - \tfrac{20}{7}R_2 = \begin{pmatrix} 7 & 1 & -5 \\ 0 & 1 & -\frac{75}{38} \\ 0 & 0 & \frac{465}{133} \end{pmatrix}. \tag{13}$$

Finally,

$$R_1 \leftarrow \tfrac{1}{7}R_1 = \begin{pmatrix} 1 & \frac{1}{7} & -\frac{5}{7} \\ 0 & 1 & -\frac{75}{38} \\ 0 & 0 & \frac{465}{133} \end{pmatrix}. \tag{14}$$

The third row is nonzero, so there are three pivots.

$$\boxed{\operatorname{rank}\begin{pmatrix} M & \mathbf{B}-\mathbf{A} \end{pmatrix} = 3, \quad \operatorname{rank}(M) = 2.} \tag{15}$$

Hence, by equation (6.1.1.4), the lines do *not* intersect.

Result

$$\boxed{\lambda = 1}, \quad \boxed{\text{The lines are perpendicular but do not intersect (they are skew).}} \tag{16}$$

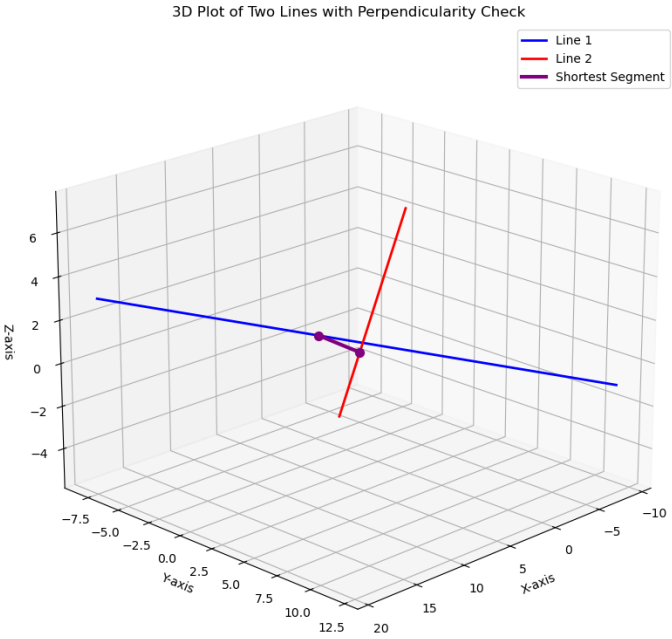

3D Plot of Two Lines with Perpendicularity Check
Line 1
Line 2
Shortest Segment
Z-axis
6
4
2
0
−2
−4
Y-axis
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5
X-axis
−10
−5
0
5
10
15
20

Fig. 1